Problem 1		Sign	Order
(i)	$4+2+1=7\equiv 1 \pmod 2$	Odd
$\operatorname{lcm}(5,3,2)=30.$		30
(ii)	$5+4=9\equiv 1 \pmod 2$	Odd
$\operatorname{lcm}(6,5)=30$		30
(iii)	$1+1+3+1=6\equiv 0 \pmod 2$	Even
$\operatorname{lcm}(2,2,4,2)=4$		4

(iv)

$$\begin{array}{c|cccccccc} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ (187\cdots) & 8 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ (123\cdots) & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{array}$$

$\Rightarrow (12345678)(18765432) = \mathrm{id}$ Even 1

Problem 2. Each permutation is a product of cycles. Thus, it suffices to show that each cycle is a product of transpositions of "$(k\ k+1)$".

To obtain $(a_1\ a_2\ \cdots\ a_k)$:

$$\begin{array}{cccc} a_1 & a_2 & \cdots & a_k \\ a_1 & a_2 & \cdots & a_k \end{array} \xrightarrow{(a_1 a_2)} \begin{array}{cccc} a_1 & a_2 & \cdots & a_k \\ a_2 & a_1 & \cdots & a_k \end{array} \xrightarrow{(a_1 a_3)} \begin{array}{ccccc} a_1 & a_2 & a_3 & \cdots & a_k \\ a_2 & a_3 & a_1 & \cdots & a_k \end{array} \xrightarrow{(a_1 a_4)} \cdots \xrightarrow{(a_1 a_k)} \checkmark$$

Conclusion : $(a_1 a_2 \cdots a_k) = (a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_3)(a_1 a_2)$.

Thus, it suffices to show that the statement holds for all transpositions (ij).

Without loss of generality, assume that $(ij) = (1\ m)$

Observation : $(j\ j{+}1)(i\ j)(j\ j{+}1) = (i\ j{+}1)$

$\Rightarrow (1\ m) = (m{-}1\ m)(1\ m{-}1)(m{-}1\ m)$

$= (m{-}1\ m)(m{-}2\ m{-}1)\cdots(2\ 3)(1\ 2)(2\ 3)\cdots(m{-}1\ m)$.

Problem 3.

(v) Yes. $Id = 0$, $(a+b\sqrt{2})^{-1} = -a-b\sqrt{2}$. Addition is associative.

$(a+b\sqrt{2})+(c+d\sqrt{2}) = (a+c)+(b+d)\sqrt{2}$. Close.

(vi)
$$\begin{pmatrix} 1 & a & b \\ & 1 & c \\ & & 1 \end{pmatrix}\begin{pmatrix} 1 & a' & b' \\ & 1 & c' \\ & & 1 \end{pmatrix} = \begin{pmatrix} 1 & a'+a & b'+ac'+b \\ & 1 & c'+c \\ & & 1 \end{pmatrix}$$

Close : ✓. Inverse : $a' = -a$, $c' = -c$, $b' = -b+ac$. ✓

Matrix multiplication is associative. Yes.

(vii) No. $(a-b)-c \neq a-(b-c)$ as long as $c \neq 0$.

(viii) Yes. Associative ✓. Close ✓.

identity : -2. inverse $a^{-1} = -a-4$.

Problem 4. "if" $h * h^{-1} = e \Rightarrow e \in H$

$e * k^{-1} = k^{-1} \Rightarrow$ close under inverse.

$h * (k^{-1})^{-1} = h * k \in H \Rightarrow$ close under multiplication.

$\Rightarrow$ H is a subgroup.

"only if" : $k \in H \Rightarrow k^{-1} \in H \Rightarrow h * k^{-1} \in H$.

Problem 5.

$ca = a \Rightarrow c = id.$

$a^2 = c = id$, $ad = f \Rightarrow af = aad = d.$

$g^2 = c = id$, $gb = d \Rightarrow gd = ggb = b.$

$bf = c = id \Rightarrow f = b^{-1} \Rightarrow fb = c$

Row 1: $\{ab, ag\} = \{b, g\}.$

$a \neq id \Rightarrow ab = g,\ ag = b.$

Column 2: $db = a$

	a	b	c	d	f	g
a	c	g	a	f	d	b
b	d	f	b	g	c	a
c	a	b	c	d	f	g
d	b	a	d	c	g	f
f	g	c	f	a	b	d
g	f	d	g	b	a	c

Row 6: $\{ga, gf\} = \{a, f\}$

$\Rightarrow ga = f,\ gf = a$

$ba = aga = af = d.$ Row 2: $\{bd, bg\} = \{a, g\} \Rightarrow bd = g,\ bg = a.$

Column 4: $fd = a$ $da = baa = bc = b.$ Column 1: $fa = g.$

Remain = Trivial.